CHM 120

Math and Graphing Prep

Final Report

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Lesson	Math and Graphing Prep
Institution	Rock Valley College
Session	Summer 2022
Course	CHM 120
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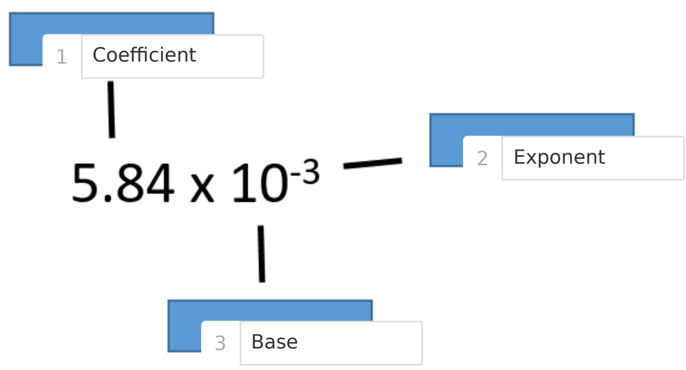
Test Your Knowledge

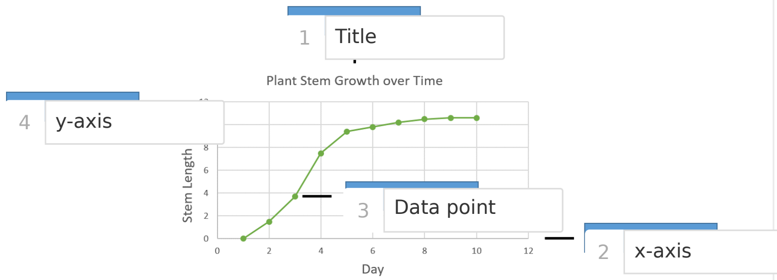
Label the components of the scientific notation.



Identify each statement as true or false.

True	False
1	2
Equation solutions should be checked by inserting the solution into the original equation.	The goal of solving equations is to isolate the known variable.
Fraction multiplication is used to solve conversion problems.	Only one factor is required to solve any unit conversion problems.

Label the parts of the graph.



Exploration

The _____ determines the how many spaces the decimal moves in scientific notation.

- ○ coefficient
- ◉ exponent
- ○ logarithm
- ○ All of the above

Dimensional analysis utilizes _____ to solve unit conversion problems.

- fraction multiplication
- conversion factors
- unit cancellation
- All of the aboe

The goal of solving an equation is to isolate the unknown variable.

- True
- Fasle

The _____ is used to represent the independent variable on a graph.

- x-axis
- y-axis
- z-axis
- All of the above

Exercise 1

Why is scientific notation often used when recording experimental measurements?

B I U 13 Word(s)

It is an easier way to write very large or very small numbers.

A scientist weighed three laboratory samples and recorded the data below. Record the total weight by determining the sum of the samples and then record the weight using scientific notation.

Sample 1: 0.00261 grams
Sample 2: 0.00362 grams
Sample 3: 0.00080 grams

B I U 2 Word(s)

0.00703

$7.03*10^{-3}$

Data Table 1: Standard Notation Conversions to Scientific Notation

Standard Notation	Scientific Notation
36000 kg	$3.6 \times 10^4 kg$
30500000 mL	$3.05 \times 10^7 mL$
0.0000936 m	$9.36 \times 10^{-5} m$
236.1 g	$2.361 \times 10^2 g$
84,300,000,000,000 mm	$8.43 \times 10^{13} mm$

Data Table 2: Scientific Notation Conversions to Standard Notation

Scientific Notation	Standard Notation
$8.13\ 10^5$ mL	$813{,}000 mL$
2.01×10^{-3} m	$0.0021 m$
9.01×10^4 mg	$90{,}100 mg$
4.48×10^{-6} L	$0.00000448 L$
5.2×10^9 mm	$5{,}200{,}000{,}000 mm$

Exercise 2

How is dimensional analysis used to convert units of measure?

B *I* U 12 Word(s)

It uses fraction multiplication where the fraction is essentially equal to 1.

A researcher combines three liquid samples with the volumes listed below. How much total liquid in liters does the researcher have? Include your calculations in your answer.

Sample 1: 250 mL
Sample 2: 1.4 L
Sample 3: 30.00 fluid ounces

B *I* U 32 Word(s)

First, convert 250mL to L: (250mL/1) * (1L/1000mL) = 250L/1000 = .25L
Second, convert 30.00 fluid oz to L: (30.00oz/1) * (29.57mL/1oz) * (1L/1000mL) = [(30.00 * 29.57 * 1)/1000] = .8871L
Add everything together: 1.4L + .25L + .8871L = 2.5371L

Answer: 2.5371L

Data Table 3: Single-Step Conversions

Conversion to Perform	Answer
32 cm to m	.32m
53.5 kg to g	53,500g
96.3 mL to L	0.0963L
4.00 miles to km	6.44km
0.25 oz to mL	7.4mL

Data Table 4: Multi-Step Conversions

Conversion to Perform	Answer
500.0 mL to qt	.528qt
0.750 yd to cm	68.28cm
0.834 gal to ml	3156.7mL
15.00 lb to kg	6.8039kg
542 in to m	13.8m

Photo 1: 32 cm to m Conversion

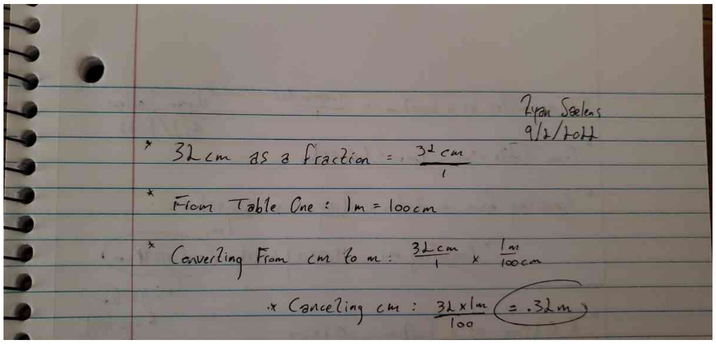


Photo 2: 53.5 kg to g Conversion

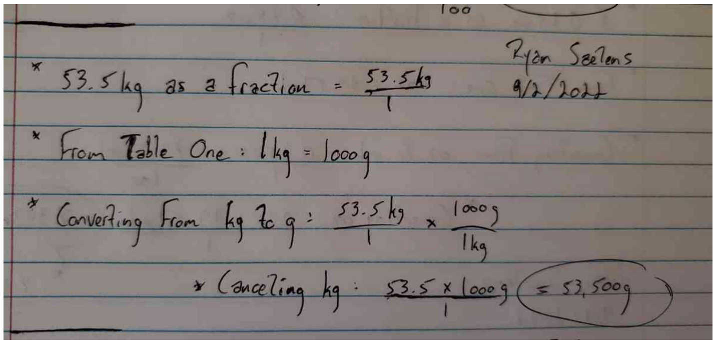
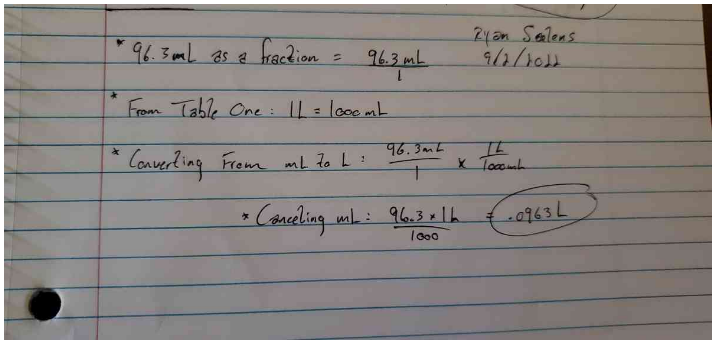


Photo 3: 96.3 mL to L Conversion

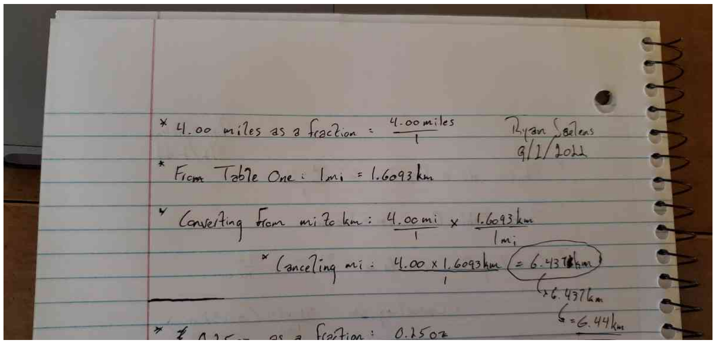


* 96.3 mL as a fraction = $\frac{96.3\text{ mL}}{1}$

Ryan Szelens
9/2/2022

* From Table One : 1L = 1000 mL
* Converting From mL to L : $\frac{96.3\text{ mL}}{1} \times \frac{1\text{ L}}{1000\text{ mL}}$
  * Canceling mL : $\frac{96.3 \times 1\text{ L}}{1000}$ = .0963 L

Photo 4: 4.00 miles to km Conversion



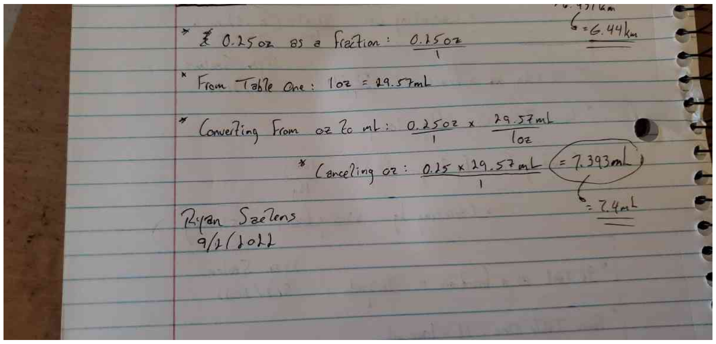
* 4.00 miles as a fraction = $\frac{4.00\text{ miles}}{1}$

Ryan Szelens
9/2/2022

* From Table One : 1 mi = 1.6093 km
* Converting from mi to km : $\frac{4.00\text{ mi}}{1} \times \frac{1.6093\text{ km}}{1\text{ mi}}$
  * Canceling mi : $\frac{4.00 \times 1.6093\text{ km}}{1}$ = 6.4372 km
  = 6.437 km
  = 6.44 km

* 0.25 oz as a fraction : 0.25 oz

Photo 5: 0.25 oz to mL Conversion



= 6.44 km

* 0.25 oz as a fraction : $\frac{0.25\text{ oz}}{1}$
* From Table One : 1 oz = 29.57 mL
* Converting From oz to mL : $\frac{0.25\text{ oz}}{1} \times \frac{29.57\text{ mL}}{1\text{ oz}}$
  * Canceling oz : $\frac{0.25 \times 29.57\text{ mL}}{1}$ = 7.393 mL
  = 7.4 mL

Ryan Szelens
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Photo 6: 500.0 mL to pt Conversion

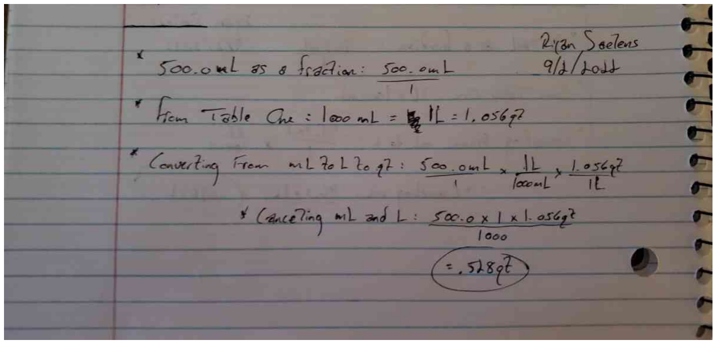

Ryan Seelens
9/2/2022
* 500.0mL as a fraction: $\frac{500.0mL}{1}$
* From Table One: 1000 mL = 1L = 1.056qt
* Converting From mL to L to qt: $\frac{500.0mL}{1} \times \frac{1L}{1000mL} \times \frac{1.056qt}{1L}$
* Canceling mL and L: $\frac{500.0 \times 1 \times 1.056qt}{1000}$
= .528qt

Photo 7: 0.750 yd to cm Conversion

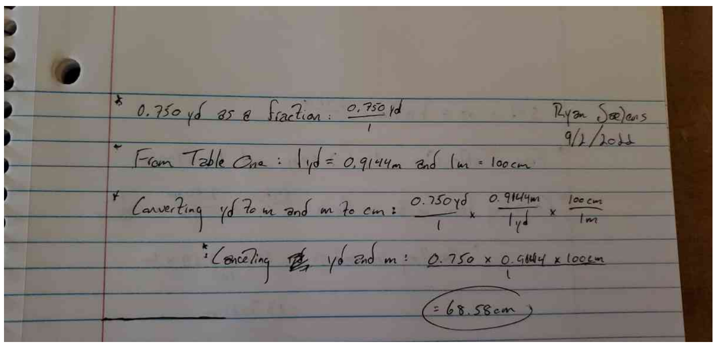

Ryan Seelens
9/2/2022
* 0.750yd as a fraction: $\frac{0.750yd}{1}$
* From Table One: 1yd = 0.9144m and 1m = 100cm
* Converting yd to m and m to cm: $\frac{0.750yd}{1} \times \frac{0.9144m}{1yd} \times \frac{100cm}{1m}$
* Canceling yd and m: $\frac{0.750 \times 0.9144 \times 100cm}{1}$
= 68.58cm

Photo 8: 0.834 gal to mL Conversion

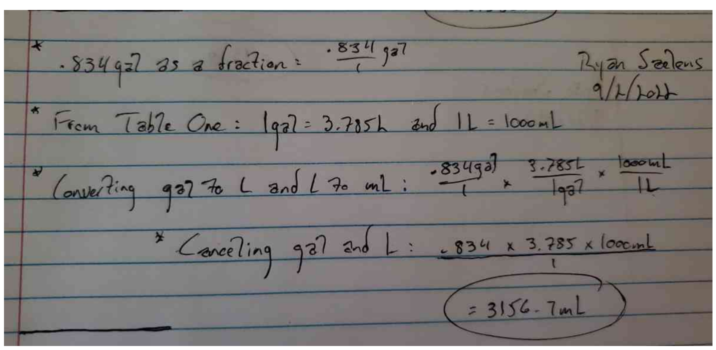

Ryan Seelens
9/2/2022
* .834gal as a fraction = $\frac{.834 gal}{1}$
* From Table One: 1gal = 3.785L and 1L = 1000mL
* Converting gal to L and L to mL: $\frac{.834gal}{1} \times \frac{3.785L}{1gal} \times \frac{1000mL}{1L}$
* Canceling gal and L: $\frac{.834 \times 3.785 \times 1000mL}{1}$
= 3156.7mL

Photo 9: 15.00 lb to kg Conversion

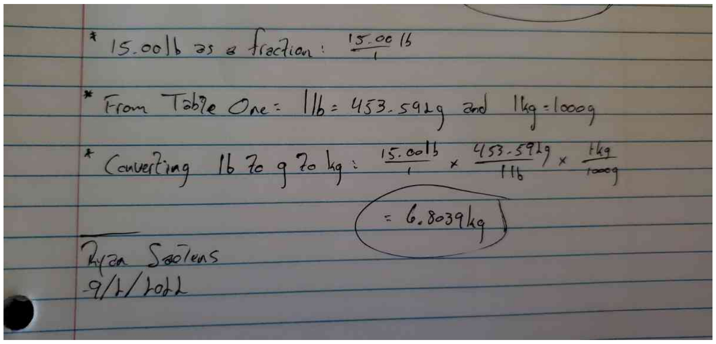
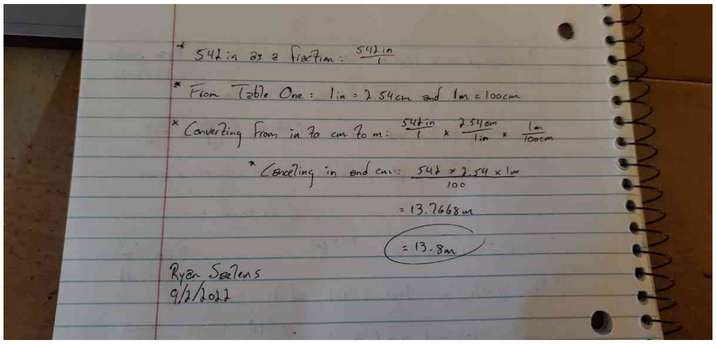


Photo 10: 542 in to m Conversion



Exercise 3

How are solutions checked after solving an equation?

B *I* U 21 Word(s)

Substitute the solution into the original equation, and if the right side is equal to the left side, everything is satisfied.

The theory of relativity is stated as the equation $E = mc^2$ where:

E = energy
m = mass
c = speed of light

A scientist has data for the speed of light and energy variables. What form of the equation should the scientist use to determine the mass of an object?

B I U 2 Word(s)

$m = E/c^2$

Data Table 5: Equation Solutions

Equation	Value of x
5x + 12 = 42	6
3x - 3 = 4	7/3
(5 + 7)/x = 24	1/2
18 = 30 - x	12
20 - 4 = 32x	1/2

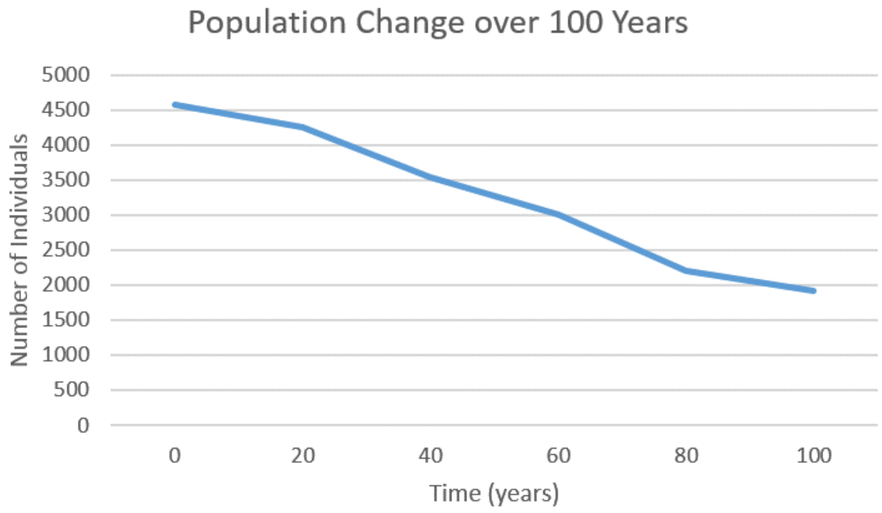
Exercise 4

How should you determine whether a bar graph or line graph is appropriate for illustrating experimental data?

B I U 39 Word(s)

If one needs to express quantities of particular variables, a bar graph is best. A line graph would be best though if one is trying to explain the relationship between variables (their dependency) , or collected data points in general.

Data for population change over 100 years for a small town is plotted on the graph below. What is the dependent variable? What trend does the graph illustrate? What was the approximate population size in year 40?



B *I* U 37 Word(s)

The dependent variable is the number of individuals. It would be strange if time was dependent upon the number of individuals.
It follows linear, negative trend.
In the year 40, the population size was roughly 3600 individuals

Data Table 6: Bacteria Numbers Dataset

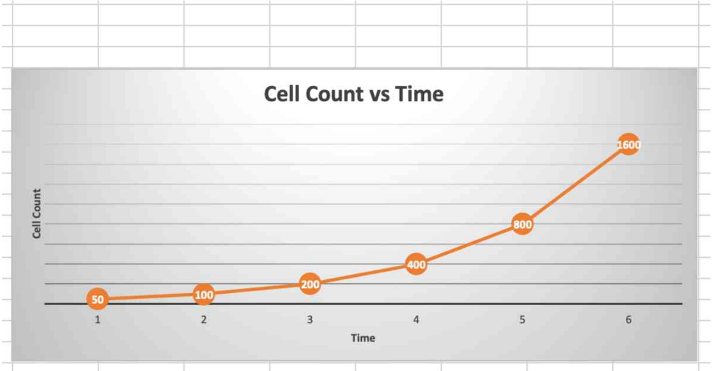
	Name	Explanation	Axis and Title
Independent Variable	Time	Time shouldn't depend on cell count. Time is fundemental and has (to a degree) always been. Cells have not 'always been'.	x-axis, Time
Dependent Variable	Cell Count	The cell count seems to increase but with respect to what...? TIME! Therefore it seems to depend on time.	y-axis, Cell Count

Panel 1: Explanation

B *I* U 28 Word(s)

A line graph is best to use for this example because we have a variable that depends on another variable. A line graph showcases their relationship quite nicely.

Graph 1: Bacteria Numbers over Time



Data Table 7: Leaf Mass Dataset

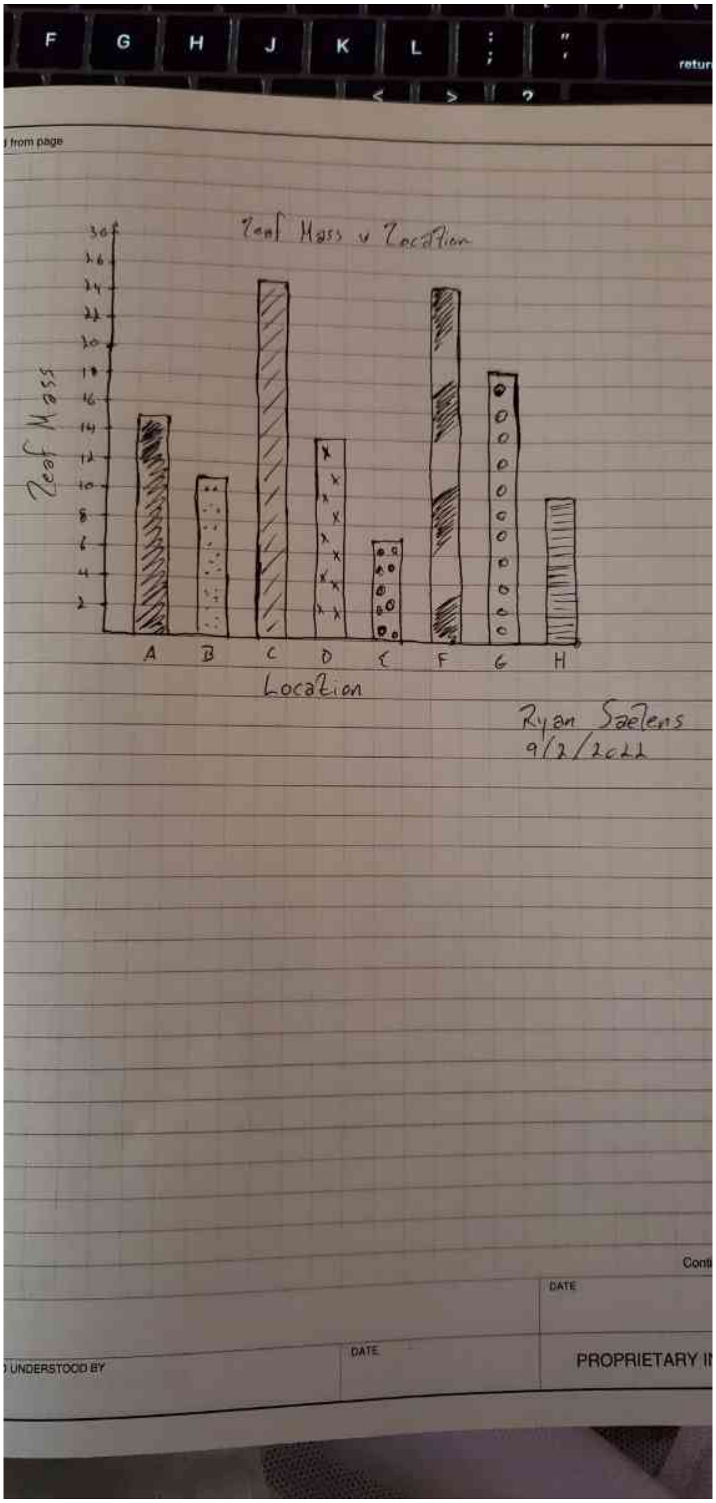
	Name	Explanation	Axis and Title
Independent Variable	Location	The leaf mass of plants relates to the location in which they are growing.	Location
Dependent Variable	Leaf Mass	Each leaf mass entry is different, and they were all grown in different locations. So the leaf mass depends on the loaction.	Leaf Mass

Panel 2: Explanation

B *I* U 28 / 10000 Word Limit

A bar graph is most appropriate here because we are showing multiple variables, each with a certain quantity. This would be hard to show with a line graph.

Graph 2: Leaf Mass per Location


Leaf Mass v Location
Leaf Mass
Location
A
B
C
D
E
F
G
H
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9/2/2022
PROPRIETARY I
DATE
UNDERSTOOD BY

Competency Review

Scientific notation is a method of writing long numbers in a(n) _____ format.

- (selected) condensed
- expanded
- unregulated
- All of the above

Negative exponents indicate numbers less than 1.00 when using scientific notation.

- (selected) True
- False

Dimensional analysis is a problem-solving technique used for _____.

- writing scientific notation
- (selected) converting units
- determining graph axes
- All of the above

_____ are created for both known measurements and conversion factors when solving unit conversion problems.

- (selected) Fractions
- Integers
- Decimals
- All of the above

Equations are solved for a(n) _____.

- known variable
- variable notation
- unknown variable
- All of the above

_____ are used to illustrate scientific data.

- Unknown variables
- Graphs
- Scientific notations
- All of the above

The y-axis of a graph is used for plotting the dependent variable.

- True
- False

The scientific notation for 236000 is _____.

- 2.30×10^6
- 2.36×10^5
- 3.62×10^7
- None of the above

Reference the conversion below to determine how many liters are in 10.0 quarts.

1 gallon = 4.00 quart

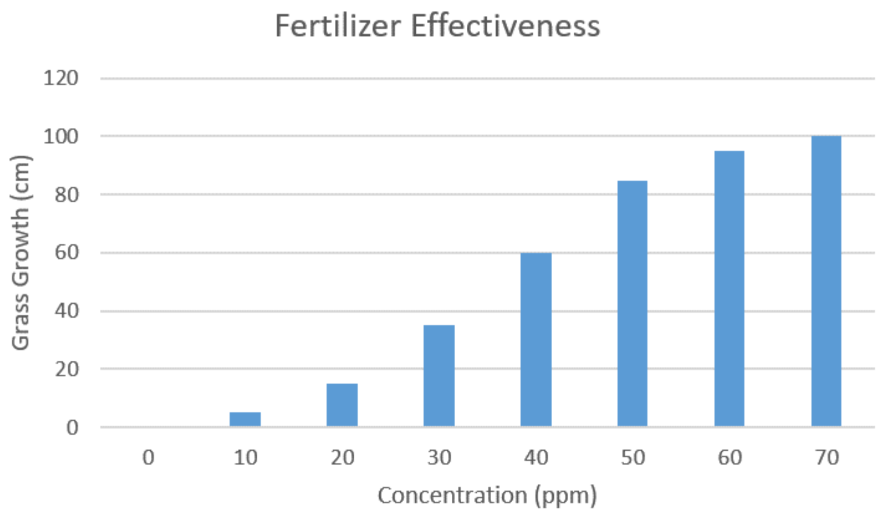
1.00 gallon = 3.78541 liters

- 9460 L
- 9.46 L
- 10.6 L
- None of the above

Solve the equation 4 + 10 = 2x.

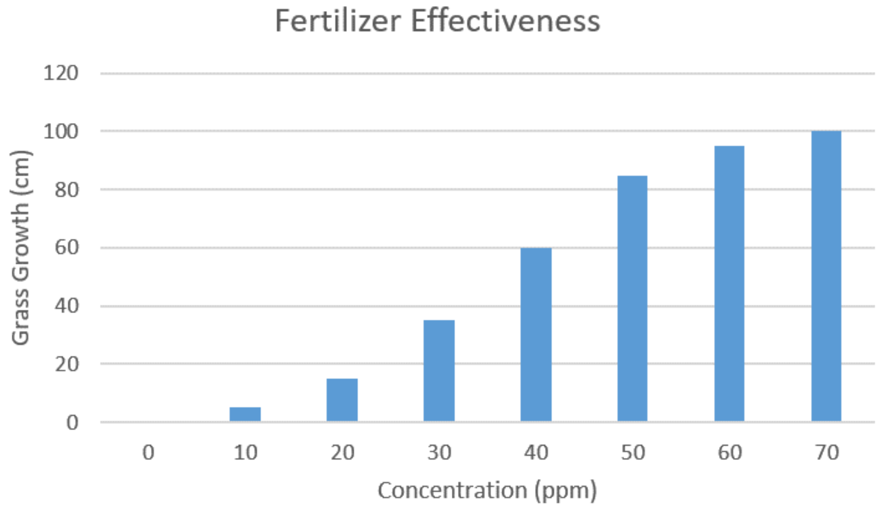
- x = 14
- x = 28
- x = 7
- None of the above

Identify the dependent variable in the graph below.



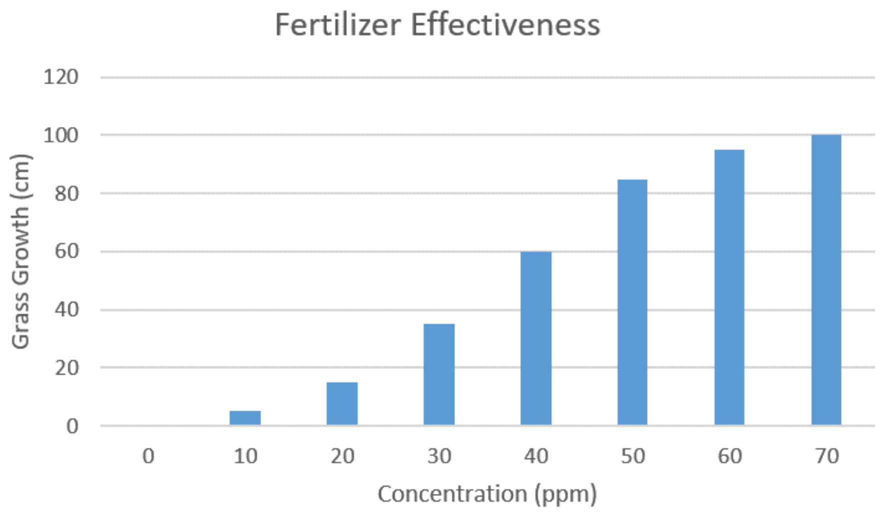
- Fertilizer effectiveness
- Concentration
- Grass growth
- None of the above

The graph below is an example of a _____.



- bar graph
- line graph
- scatter plot
- None of the above

The graph below illustrates that increases in concentration result in _____ in grass growth.



- decreases
- increases
- no changes
- None of the above

Extension Questions

Dr. Jones performed an experiment to monitor the effects of sunlight exposure on stem density in aquatic plants. In the study, Dr. Jones measured the mass and volume of stems grown in 5 levels of sun exposure. The data is represented below.

Sun exposure (%)	Stem mass (g)	Stem volume (mL)
30	275	1100
45	415	1215
60	563	1425
75	815	1610
90	954	1742

a. Convert the mass measurements to kilograms (kg) and the volume measurements to cubic meters (m^3).

b. Calculate the density of the samples using the equation d = m/v.

d = density

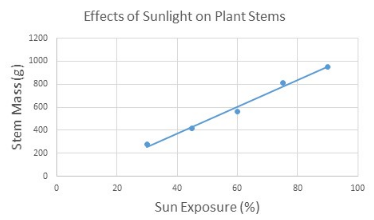
m = mass (kg)

v = volume (m^3)

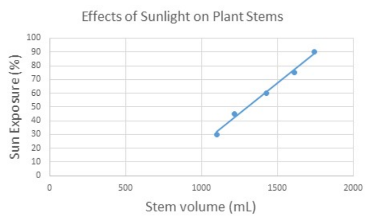
c. Convert the density values to scientific notation.

d. Select the graph that best represents the data.

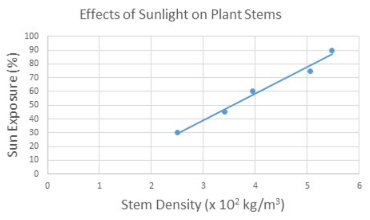
A



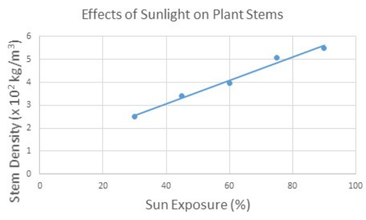
B



C



D



C